(Please write your Enrollment Number) Enrollment No._____

END-TERM EXAMINATION

B.Tech. CSE-AI / AI&ML Semester: III

(November-December, 2023) OFFLINE MODE

Subject Code: BAI-201	Subject: ARTIFICIAL INTELLIGENCE

Time: 3 Hours **Maximum Marks:** 60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(2.5*8)**	
	(a) Differentiate between Strong AI and Weak AI in the form of table		
	(b) What do you mean by Heuristic and Heuristic function? Explain		
	(c) What do you mean by feature extraction? Briefly explain popular methods of feature extraction.		
	(d) Differentiate between rule-based and learning-based AI approaches		
	(e) Explain membership function of a fuzzy set? Give examples		
	(f) Define Bayesian belief network, and state applications of this network		
	(g) Explain any four applications of IoT		
	(h) Explain Metasearch Engine, and state it's pros and cons		
	UNIT-I		
Q2	**Answer the questions in part I) and II) based on the figure 1 below:**	**(10)**	

Please refer to Figure 1 and answer the following questions.

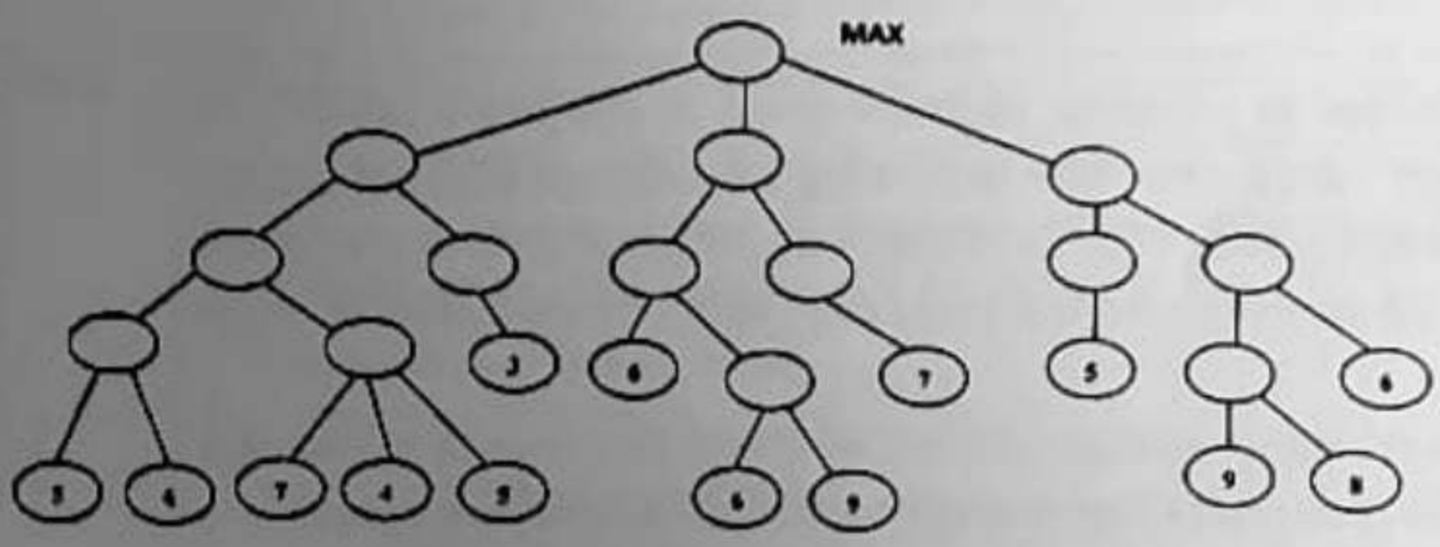


Figure 1

Apply the **minimax** algorithm to the game tree in Figure 1, where it is the MAXIMIZER's turn to play. The values estimated by the evaluation function are indicated at the leaf nodes. Assume that the search always visits children in a left-to-right order.

i) Compute the backed-up values computed by the **minimax** algorithm. Show your answer by writing values at the appropriate nodes in the above tree.

ii) Indicate the proper move for the MAXIMIZER by marking the line with bold indication(⟶) to the root's outgoing edges.

Q3	What is minimax algorithm in A1? Explain its steps? What is contingency problem in Minimax? State advantages and disadvantages of Minimax?	(10)	
	UNIT-II		
Q4	Explain Word embeddings in the context of AI? Where are they used? Explain different approaches to get Word embeddings, and list the properties of different approaches of Word embeddings?	(10)	
Q5	Consider these statements and answer the questions below: (a) Convert all the above statements into predicate logic (b) Show that John likes peanuts using back chaining (c) Convert the statements into clause form (d) Using Resolution show that "John likes peanuts" **Statements:** John likes all kinds of food. Apples are food. Chicken is food. Anything anyone eats and it is not killed is food. Bill eats peanuts and is still alive. Swe eats everything Bill eats	(10)	
	UNIT-III		
Q6	a) A doctor has decided to prescribe two new drugs to 200 heart patients as follows: 50 get drug A, 50 get drug B and 100 get both. Drug A reduces the probability of a heart attack by 35 percent, drug B reduces the probability by 20 percent, and the two drugs when taken together, work independently. The 200 patients were chosen so that each has an 80 percent chance of having a heart attack, find the probability that the patient was given both the drugs? b) Consider fuzzy set A defined on the interval x= [0,10] of integers by the membership function $\mu A(x) = x / (x+ 3)$. Find α cut corresponding to $\alpha = 0.6$	(10)	
Q7	How do AI-based systems deal with Uncertainty? State the strategies adopted to handle Uncertainty problems in AI? Explain Bayesian belief networks and their applications?	(10)	
	UNIT-IV		
Q8	Explain the Importance of Natural Language Processing (NLP) and its components? Explain NLP techniques and any five applications of NLP.	(10)	
Q9	Explain the significance and scope of Expert Systems in AI, and the components of an Expert System? Elucidate the steps for development of expert system along with its participants?	(10)	